SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Salt 1 Preliminary Test		
1)	Note the colour of the given salt.	Colourless	Absence of Cu^{2+} Ni^{2+}, Mn^{2+}, Fe^{2+}
2)	Note the smell of the given salt.	No characteristic smell	Absence of NH_4^+, CH_3COO^-
3)	Solubility in dil. HCl	Soluble	Absence of Pb^{2+}
4)	Flame test A pinch of salt is made into paste with conc. HCl and the paste is introduced to the flame by using a glass rod.	Brick red flame	Presence of Ca^{2+}
5)	Ash test	No characteristic	Absence of Al^{3+},

A pinch of salt is made into paste with conc. HCl and the paste is introduced to the flame by using a glass rod.	Brick red flame	Presence of [illegible]

5) Ash test The test solution is mixed with few drops of conc. HNO_3 and a little cobalt nitrate solution. A rolled filter paper dipped in to it is burnt to ash.	No characteristic colour in the ash	Absence of Al^{3+}, Zn^{2+}, Mg^{2+}
Analysis of anion		
1) To the test solution add dil. HCl.	Brisk efforvesence of colourless, odourless, gas which turns lime water milky.	Presence of carbonate (CO_3^{2-})

$$M_2CO_3 + 2HCl \rightarrow 2MCl + CO_2 + H_2O$$

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
2)	The test solution is rubbed with dil. H_2SO_4	No smell of vinegar	Absence of acetate (CH_3COO^-)
3)	To a little of the salt solution about 1ml of conc. H_2SO_4 is added & heated.	No dense white gas.	Absence of chloride (Cl^-)
4)	A little of the salt soln is heated with conc. H_2SO_4 & a paper ball is added	No reddish brown fumes	Absence of nitrate (NO_3^-)
5)	To the test solution add $BaCl_2$ solution.	No white precipitate.	Absence of sulphate (SO_4^{2-})

Confirmatory Test

$BaCl_2$ solution.		sulphate (SO_4^{2-})

Confirmatory Test

Carbonate

1) To the test solution add $BaCl_2$ solution	white precipitate which is soluble in dil. HCl.	Presence of CO_3^{2-} confirmed

$$M_2CO_3 + BaCl_2 \rightarrow 2MCl + BaCO_3$$

$$BaCO_3 + 4HCl \rightarrow BaCl_2 + CO_2 + H_2O$$

2) A little of the solid salt is heated with dil. H_2SO_4	Colourless, odourless gas which turns lime water milky.	Presence of CO_3^{2-} confirmed.

$$CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2\uparrow$$

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Analysis of cations Intergroup seperation		
1)	To the test solution add dil Na_2CO_3	white precipitate	Absence of 0 group (NH_4^+)
2)	To the test solution add dil. HCl	No white precipitate	Absence of I group (Pb^{2+}).
3)	To the test solution add dil. HCl, & pass H_2S gas.	No black precipitate	Absence of II group (Cu^{2+})
4)	To the test solution add dil. HCl, NH_4Cl & NH_4OH pass H_2S gas.	No white gelatinous precipitate.	Absence of III group (Al^{3+}, Fe^{2+})
5)	To the test soln add dil. HCl, NH_4Cl & NH_4OH pass H_2S gas.	No precipitate	Absence of IV group (Zn^{2+}, Mn^{2+})
6)	To the test soln add dil. HCl, NH_4Cl, NH_4OH	white precipitate	Presence of V group (Ca^{2+}, Ba^{2+},

4) To the test solution add dil. HCl, NH_4Cl & NH_4OH pass H_2S gas.	No white gelatinous precipitate	Absence of III group (Al^{3+}, Fe^{2+})
5) To the test soln add dil. HCl, NH_4Cl & NH_4OH pass H_2S gas.	No precipitate	Absence of IV group (Zn^{2+}, Mn^{2+})
6) To the test soln add dil HCl, NH_4Cl, NH_4OH and $(NH_4)_2CO_3$	white precipitate	Presence of V group (Ca^{2+}, Ba^{2+}, Sr^{2+})
7) To the test soln add dil. HCl, NH_4Cl, NH_4OH & Na_2HPO_4	No white precipitate	Absence of VI group (Mg^{2+})

Confirmatory test for fifth group (Ca^{2+}, Ba^{2+}, Sr^{2+})

1) To the test soln acetic acid & potassium chromate soln is added	Yellow solution	Presence of Ca^{2+}

$$Ca^{2+} + K_2C_2O_4 \longrightarrow CaC_2O_4 + 2K^+$$

2) Ammonium oxalate solution and NH_4OH_4 soln is added to the test soln. To the above precipitate acetic acid is added & heated.	white precipitate insoluble in acetic acid	Presence of Ca^{2+}

$$Ca^{2+} + (NH_4)_2C_2O_4 \longrightarrow CaC_2O_4 + 2NH_4^+$$

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Salt 2		
	Preliminary Test		
1.	Noted the colour of the given salt	Colourless	Absence of Cu^{2+}, Ni^{2+}, Mn^{2+}, Fe^{2+}
2.	Noted the smell of the given salt.	No characteristic smell.	Absence of NH_4^+, CH_3COO^-
3.	Solubility in dil. HCl.	Insoluble	Presence of Pb^{2+}
4.	Flame test A pinch of salt solution is made to paste with conc. HCl and it is introduced to flame by using a glass rod.	No characteristic colour on flame.	Absence of Ca^{2+}, Sr^{2+}, Ba^{2+}
5.	Ash test The test solution is	No characteristic colour on ash	Absence of Al^{3+}, Zn^{2+}, Mg^{2+}.

4.	Flame test A pinch of salt solution is made to paste with conc. HCl and it is introduced to flame by using a glass rod.	No characteristic colour on flame.	Absence of Ca^{2+}, Sr^{2+}, Ba^{2+}

5.	Ash test The test solution is mixed with few drops of conc. HNO_3 and a little cobalt nitrate solution. A rolled filter paper dipped in to it is burnt to ash.	No characteristic colour on ash	Absence of Al^{3+}, Zn^{2+}, Mg^{2+}.
	Analysis of anion		
1.	To the test solution add dil HCl.	No Brisk effervence.	Absence of carbonate (CO_3^{2-}).
2.	To the test solution add dil. H_2SO_4	Smell of vinegar	Presence of acetate (CH_3COO^-).
	$H_2SO_4 + 2CH_3COOH \longrightarrow 2CH_3COOH + H_2SO_4$		
3.	To the little of the salt solution is heated with conc. H_2SO_4 and a paper ball	No reddish brown fumes	Absence of nitrate (NO_3^-)

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	is added.		
4.	To the little of the salt solution about 1ml of conc. H_2SO_4 is added & heated.	No dense white gas.	Absence of chloride (Cl^-).
5.	To the test solution add $BaCl_2$ solution.	No white precipitate	Absence of sulphate (SO_4^{2-})
	Confirmatory test		
	Acetate		
1.	To the test solution add natural ferric chloride solution.	Deep red colouration	Presence of acetate.

$$3CH_3COOH + FeCl_3 \longrightarrow (CH_3COO)_3Fe + 3HCl$$

2.	To the test solution add ethanol and 1ml conc. H_2SO_4.	Pleasant fruity.	Presence of acetate.

Confirmatory test
Acetate

1.	To the test solution add neutral ferric chloride solution.	Deep red colouration	Presence of acetate.

$$3CH_3COOH + FeCl_3 \longrightarrow (CH_3COO)_3Fe + 3HCl$$

2.	To the tpt solution add ethanol and 1ml conc. H_2SO_4. Heat the mixture and pour the contents into a beaker containing water.	Pleasant fruity.	Presence of acetate

$$2CH_3COONa + H_2SO_4 \longrightarrow Na_2SO_4 + 2CH_3COOH$$
$$CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$$
(Fruity smell)

Analysis of cation intergroup seperation

1.	To the test solution add dil. Na_2CO_3	white precipitate	Absence of 0 group (NH_4^+)
2.	To the test solution add dil. HCl.	white precipitate.	Presence of I group (Pb^{2+})
3.	To the test solution add dil. HCl and pass H_2S gas.	No Black precipitate	Absence of II group (Cu^{2+})

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
4.	To the test solution add dil HCl, NH_4Cl and NH_4OH.	No white precipitate.	Absence of III group (Al^{3+}, Fe^{2+})
5	To the test solution add dil. HCl, NH_4Cl and NH_4OH pass H_2S gas.	No white & flesh colourprecipitate.	Absence of IV group (Ca^{2+}, Ba^{2+}, Sr^{2+})
6.	To the test solution add dil. HCl, NH_4Cl, NH_4OH and $(NH_4)_2CO_3$.	No white precipitate	Absence of V group (Zn^{2+}, Mn^{2+})
7.	To the test solution add dil HCl, NH_4Cl, NH_4OH and Na_2HPO_4.	No white precipitate.	Absence of VI group (Mg^{2+}).

Confirmatory test for lead ion (Pb^{2+}).

1)	To the test solution is acidified with acetic	Yellow precipitate	Presence of Pb^{2+}

6.	To the test solution add dil HCl, NH_4Cl, NH_4OH and $(NH_4)_2CO_3$.	No white precipitate	Absence of V group (Zn^{2+}, Mn^{2+})
7.	To the test solution add dil HCl, NH_4Cl, NH_4OH and Na_2HPO_4.	No white precipitate.	Absence of VI group (Mg^{2+}).

Confirmatory test for lead ion (Pb^{2+}).

1)	To the test solution is acidified with acetic acid & potassium chromate is added	Yellow precipitate	Presence of Pb^{2+}

$$Pb^{2+} + K_2CrO_4 \longrightarrow PbCrO_4 + 2K^+$$

2)	Potassium iodide solution is added to the test solution	Yellow precipitate	Presence of lead

$$Pb^{2+} + K_2CrO_4 \longrightarrow PbI_2\downarrow + 2K^+$$

The above precipitate is boiled with water & cooled.	Precipitate dissolves on boiling Golden yellow sparkles is found on cooling.	Presence of lead confirmed.

Result

The given salt contain lead as cation & acetate as anion. Therefore the salt is lead acetate.

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SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Salt 3 Preliminary Test		
1.	Note the colour of the given salt.	Colourless	Absence of Cu^{2+} Ni^{2+}, Mn^{2+}, Fe^{2+}
2.	Note the smell of the given salt.	Ammonial smell	May be the presence of NH_4^+
3.	Solubility in dil. HCl	Soluble	Absence of Pb^{2+}
4.	Flame test A pinch of salt is made into paste with conc. HCl and the paste is introduced to the flame by using a glass rod.	No characteristic colour on the flame.	Absence of Ca^{2+} Sr^{2+}, Ba^{2+}
5.	Ash test The solution is mixed	No characteristic colour on the	Absence of Al^{3+}, Zn^{2+}, Mg^{2+}

	into paste with conc. HCl and the paste is introduced to the flame by using a glass rod.	flame.	
5.	Ash test The solution is mixed with few drops of conc. HNO_3 and a little cobalt nitrate soln. A rolled filter paper dipped in to it is burnt to ash.	No characteristic colour on the ash.	Absence of Al^{3+}, Zn^{2+}, Mg^{2+}
	Analysis of Anion		
1.	To the test solution add dil. HCl.	No Brisk effervescence.	Absence of carbonate (CO_3^{2-}).
2.	The test solution is rubbed with dil. H_2SO_4.	No vinegar like smell.	Absence of acetate (CH_3COO^-)
3.	To a little of the salt soln about 1ml of conc. H_2SO_4 is added and heated	Dense white gas which fumes with NH_4OH solution taken on a glass rod	Presence of chloride (Cl^-)

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	$MCl + H_2SO_4 \longrightarrow MHSO_4 + HCl$ $HCl + NH_4OH \longrightarrow NH_4Cl + H_2O$		
4.	A little of the salt soln is heated with conc. H_2SO_4 and a paper ball is added.	No Reddish brown fumes	Absence of nitrate (NO_3^-).
5.	To the test solution add $BaCl_4$ soln	No white precipitate	Absence of sulphate (SO_4^{2-})
6	<u>Confirmatory Test</u>.		
	<u>Chloride</u>		
1.	To the test solution is heated with conc H_2SO_4 and a pinch of manganese dioxide (MnO_2).	Greenish yellow gas with irritating smell.	Presence of chloride.
	$2MCl + MnO_2 + 3H_2SO_4 \longrightarrow 2MHSO_4 + MnSO_4 + Cl_2 + 2H_2O$		
2.	The test solution is mixed with dil HNO_3	White curdy precipitate soluble	Presence of chlorine confirmed

5.	To the test solution add $BaCl_4$ soln	No white precipitate	Absence of sulphate (SO_4^{2-})
6	**Confirmatory Test.**		
	Chloride		
1.	To the test solution is heated with conc H_2SO_4 and a pinch of manganese dioxide (MnO_2)	Greenish yellow gas with irritating smell	Presence of chloride.
	$2HCl + MnO_2 + 3H_2SO_4 \longrightarrow 2MHSO_4 + MnSO_4 + Cl_2 + 2H_2O$		
2	The test solution is mixed with dil. HNO_3 and add $AgNO_3$ solution	White curdy precipitate soluble in excess of NH_4OH	Presence of chlorine confirmed.
	$HCl + AgNO_3 \longrightarrow AgCl + HNO_3$ $AgCl + 2NH_4OH \longrightarrow [Ag[NH_3]_2]Cl + 2H_2O$		
	Analysis of cation (Intergroup seperation)		
i)	To the test soln add dil. Na_2CO_3	No precipitate	Presence of 0 group (NH_4^+)
2)	To the test soln add dil. HCl	No white precipitate	Absence of I group (Pb^{2+}).
3)	To the test soln add dil HCl, NH_4Cl, NH_4OH.	No black precipitate	Absence of II group (Cu^{2+})
4)	To the test soln add dil HCl, NH_4Cl, NH_4OH	No white gelatinous precipitate	Absence of III group (Al^{3+}, Fe^{2+})

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
5)	To the test soln add dil. HCl, NH_4Cl and NH_4OH gas H_2S.	No white precipitate.	Absence of IV group (Mn^{2+}, Zn^{2+})
6)	To the test soln add dil HCl, NH_4Cl and NH_4OH and $(NH_4)_2CO_3$.	No white precipitate.	Absence of V group (Ca^{2+}, Ba^{2+}, Sr^{2+})
7)	To the test solution add dil HCl, NH_4Cl, NH_4OH and Na_2HPO_4	No white precipitate	Absence of VI group (Mg^{2+}).
	Confirmatory test for	ammonium ion	
1)	NaOH soln is added the test solution is excess	Colourless gas with smell of ammonia The gas gives white fume when a glass rod dipped in dil.	Presence of NH_4^+

…, NH_4Cl, NH_4OH and Na_2HPO_4	precipitate	… (Hg^{2+}).

Conformatory test for ammonium ions

i) NaOH soln is added the test solution in excess	Colourless gas with smell of ammonia … when a glass rod dipped in dil. HCl is brought near the mouth.	Presence of NH_4^+

$$NH_4^+ + NaOH \longrightarrow Na^+ + H_2O + NH_3\uparrow$$

$$NH_3 + HCl \longrightarrow NH_4Cl$$

ii) Nessler's reagent is added to the test soln.	Brown precipitate	Presence of NH_4^+

$$2K_2[HgI_4] + NH_3 + 3KOH \longrightarrow H_2N\cdot HgO\cdot HgI + 7KI + 2H_2O$$

Result

The result salt contain ammonium as cation and chloride as anion.

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SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Salt 4		
1.	Note the colour of the given salt.	Colourless	Absence of Cu^{2+}, Ni^{2+}, Mn^{2+}, Fe^{2+}
2.	Note the smell of the given salt.	No characteristic smell	Absence of NH_4^+, CH_3COO^-
3.	Solubility in dil. HCl	Soluble	Absence of Pb^{2+}
4.	Flame test A pinch of salt is made into paste with conc. HCl & the paste is introduced to the flame by using a glass rod.	NO characteristic colour on the flame	Absence of Ca^{2+}, Sr^{2+}, Ba^{2+}
5.	Ash test The test solution is mixed with few drops of conc. HNO_3 and a little cobalt nitrate solution A rolled filter paper	Green tinted ash	Presence of Zinc

	Experiment	Observation	Inference
4.	Flame test A pinch of salt is made into paste with conc. HCl & the paste is introduced to the flame by using a glass rod.	No characteristic colour on the flame	Absence of Ca^{2+}, Sr^{2+}, Ba^{2+}
5.	Ash test The test solution is mixed with few drops of conc. HNO_3 and a little cobalt nitrate solution. A rolled filter paper dipped in to it is burnt to ash.	Green tinted ash	Presence of Zinc.

Analysis of anion

1.	To the test solution add dil. HCl	No Brisk effervescence.	Absence of carbonate (CO_3^{2-})
2.	To the test solution is rubbed with dil. H_2SO_4	No smell of vinegar.	Absence of acetate (CH_3COO^-)
3.	To a little of the salt soln about 1ml of conc. H_2SO_4 is added & heated.	No dense white gas	Absence of chloride (Cl^-)
4.	A little of the salt soln is heated with conc. H_2SO_4 and paper ball is added.	No reddish brown fumes.	Absence of nitrate (NO_3^-)

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
5.	To the test solution add $BaCl_2$ solution	white precipitate which is insoluble in dil. HCl	Presence of sulphate (SO_4^{2-})

$$M_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2MCl$$

Confirmatory Test
Sulphate

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
1.	To the test soln add lead acetate	white precipitate	Presence of SO_4^{2-}

$$M_2SO_4 + Pb(CH_3COO)_2 \longrightarrow PbSO_4 + 2CH_3COOM$$

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
2.	To the above precipitate add ammonium acetate solution & heat it.	white precipitate dissolves	Presence of SO_4^{2-}

$$PbSO_4 + 2CH_3COONH_4 \longrightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$$

Analysis of cation intergroup separation

2.	To the above precipitate add ammonium acetate solution & heat it.	white precipitate dissolves	Presence of SO_4^{2-}

$$PbSO_4 + 2CH_3COONH_4 \longrightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$$

Analysis of cation intergroup separation

1.	To the test soln add dil. Na_2CO_3	white precipitate	Absence of 0 group (NH_4^+).
2.	To the test solution add dil. HCl.	No white precipitate.	Absence of I group (Pb^{2+}).
3.	To the test solution add dil. HCl & pass H_2S gas.	No black precipitate.	Absence of II group (Cu^{2+}).
4.	To the test solution add dil HCl, NH_4Cl & NH_4OH.	No white gelatinous precipitate.	Absence of III group (Al^{3+}, Fe^{2+})
5.	To the test soln add dil. HCl, NH_4Cl, NH_4OH & Pass H_2S gas.	white precipitate	Presence of IV group (Zn^{2+}, Mn^{2+})
6.	To the test solution add dil. HCl, NH_4Cl, NH_4OH and $(NH_4)_2CO_3$	No white precipitate.	Absence of V group (Ca^{2+}, Ba^{2+}, Sr^{2+})

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
7.	To the test solution add dil. HCl, NH_4Cl, NH_4OH and Na_2HCO_4	No white precipitate	Absence of VI group (Mg^{2+})
	Confirmatory test for Zinc ion		
1)	Potassium ferrocyanide solution is added to the test solution	white precipitate.	Presence of Zn^{2+}.
	$2Zn^{2+} + K_4[Fe(CN)_6] \rightarrow Zn_2[Fe(CN)_6] + 4K^+$ white or bluish white precipitate.		
2)	Ash test The test solution is mixed with few drops of conc. HNO_3 and a little cobalt nitrate solution.	Green tinted ash.	Presence of Zn^{2+} confirmed.

A rolled filter paper dipped in to it is burnt to ash.			

the test solution

$2Zn^{2+} + K_4[Fe(CN)_6] \rightarrow Zn_2[Fe(CN)_6] + 4K^+$
white or bluish white precipitate.

2)	Ash test The test solution is mixed with few drops of conc. HNO_3 and a little cobalt nitrate solution.	Green tinted ash.	Presence of Zn^{2+} confirmed.
	A rolled filter paper dipped in to it is burnt to ash.		

$$2Co(NO_3)_2 \xrightarrow{\Delta} 2CoO + 4NO_2 + O_2$$
$$ZnO + CoO \rightarrow ZnO \cdot CoO \text{ (green)}$$

Result

The given salt contain zinc as cation & sulphate as anion. Therefore the salt is Zinc Sulphate.

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SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Salt 5		
1.	Note the colour of the given salt	Colourless	Absence of Cu^{2+}, Ni^{2+}, Mn^{2+}, Fe^{2+}.
2.	Note the smell of the given salt.	No characteristic smell	Absence of NH_4^+, CH_3COO^-
3.	Solubility in dil. HCl	Insoluble	Presence of Pb^{2+}
4.	Flame test A pinch of salt is made into paste with conc. HCl & the paste is introduced into the flame by using a glass rod.	No characteristic colour on the flame.	Absence of Ca^{2+}, Ba^{2+}, Sr^{2+}.

4.	Flame test A pinch of salt is made into paste with conc. HCl & the paste is introduced into the flame by using a glass rod.	No characteristic colour on the flame.	Absence of Ca, Ba^{2+}, Sr^{2+}.
5.	Ash test The test solution is mixed with few drops of conc. HNO_3 and a little cobalt nitrate solution. A rolled filter paper dipped in to it is burnt to ash.	No characteristic colour on the ash.	Absence of Al^{3+}, Zn^{2+}, Mg^{2+}.
	Analysis of anion		
1)	To the test soln add dil. HCl	No Brisk effervescence	Absence of carbonate.
2)	The test solution is rubbed with dil. H_2SO_4	No smell of vinegar.	Absence of acetate.
3)	To the little of the salt soln about 1ml of conc. H_2SO_4 is added & heated.	No dense white gas.	Absence of chloride.

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
4)	To the test soln add $BaCl_2$ soln	No white precipitate	Absence of sulphate.
5)	A little of the salt solution is heated with conc. H_2SO_4 & a paper ball is added.	Reddish brown fumes.	Presence of nitrate (NO_3^-).

$$MNO_3 + H_2SO_4 \longrightarrow MHSO_4 + HNO_3$$
$$4HNO_3 + \underset{\text{(paper ball)}}{C} \longrightarrow 4NO_2\uparrow + CO_2 + 2H_2O$$

Confirmatory test

Nitrate

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
1.	The test solution is mixed with freshly prepared ferrous sulphate solution Con. H_2SO_4 is then added slowly through the sides of the test tube.	A dark brown ring is formed at the junction of two liquids.	Presence of nitrate.

(paper ball)

Confirmatory test Nitrate 1. The test solution is mixed with freshly prepared ferrous sulphate solution Conc H_2SO_4 is then added slowly through the sides of the test tube.	A dark brown ring is formed at the junction of two liquids.	Presence of nitrate.

$$MNO_3 + H_2SO_4 \longrightarrow MHSO_4 + HNO_3$$
$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \longrightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO$$
$$FeSO_4 + NO + 5H_2O \longrightarrow [FeNO(H_2O)_5]SO_4$$

(Brown ring)

2. Small quantity of solid salt is heated with conc. H_2SO_4 & few copper chips.	Reddish brown fumes	Presence of nitrate confirmed.

$$2MNO_3 + H_2SO_4 \longrightarrow M_2SO_4 + 2HNO_3$$
$$4HNO_3 + Cu \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

(Reddish brown fumes)

Analysis of cation Intergroup Separation

To the test solution add dil Na_2CO_3.	white precipitate	Absence of 0 group (NH_4^+)

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
2.	To the test soln add dil. HCl.	white precipitate	Presence of I group (Pb^{2+})
3.	To the test soln add dil. HCl & pass H_2S gas.	No Black precipitate	Absence of II group (Cu^{2+}).
4.	To the test solution add dil HCl, NH_4Cl and NH_4OH & pass H_2S.	No white precipitate	Absence of III group (Al^{3+}, Fe^{2+}).
5.	To the test solution add dil HCl, NH_4Cl & NH_4OH & $(NH_4)_2CO_3$.	No white precipitate.	Absence of IV group (Zn^{2+}, Mn^{2+}).
6.	To the test solution add dil. HCl, NH_4Cl & NH_4OH & $(NH_4)_2CO_3$	No white precipitate	Absence of V group (Ca^{2+}, Ba^{2+}, Sr^{2+})
7.	To the test solution add dil. HCl, NH_4Cl, NH_4OH &	No white precipitate.	Absence of VI group (Mg^{2+})

6. To the test solution add dil. HCl, NH_4Cl & NH_4OH & $(NH_4)_2CO_3$	No white precipitate	Absence of V group (Ca^{2+}, Ba^{2+}, Sr^{2+})

7. To the test solution add dil. HCl, NH_4Cl, NH_4OH & Na_2HPO_4.	No white precipitate.	Absence of VI group (Mg^{2+})

Confirmatory test for lead ion (Pb^{2+})

1) The test solution is acidified with acetic & potassium chromate is added	Yellow precipitate.	Presence of Pb^{2+}
$Pb^{2+} + K_2CrOH \longrightarrow PbCrO_4\downarrow + 2K^+$		
2) Potassium iodide soln is added to the test soln.	Yellow precipitate	Presence of Pb^{2+}
	$Pb^{2+} + 2KI \longrightarrow PbI_2\downarrow + 2K^+$	
The above precipitate is boiled with water & cooled.	Precipitate dissolves in boiling Golden yellow spangles is formed on cooling.	Presence of Pb^{2+} confirmed.

Result

The given salt contain lead as cation & nitrate as anion ∴ The salt is lead Nitrate

25/03

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Salt 6		
	Preliminary test		
1.	Note the colour of the given salt.	Colourless.	Absence of Cu^{2+}, Ni^{2+}, Mn^{2+}, Fe^{2+}
2.	Note the smell of the given salt.	No characteristic smell.	Absence of NH_4^+ and CH_3COO^-.
3.	Solubility in dil. HCl	Soluble	Absence of Pb^{2+}.
4.	Flame test A pinch of salt is made into paste with conc. HCl & the paste is introduced to the flame by using a glass rod.	No characteristic colour on the flame.	Absence of Ca^{2+}, Sr^{2+}, Ba^{2+}.

5)	Ash test	Blue tinted	Presence of

... of the paste is introduced to the flame by using a glass rod.	flame.	

7) Ash test The soln is mixed with few drops of conc HNO_3 & cobalt nitrate. A rolled filter paper dipped & burnt to ash.	Blue tinted ash.	Presence of Al^{3+}
Analysis of anion		
1) To the test soln add dil. HCl	No Brisk effervescence	Absence of CO_3^{2-}
2) To the test soln is rubbed with dil. H_2SO_4.	No smell of vinegar	Absence of CH_3COO^-.
3) To a little of salt soln about 1ml of conc. H_2SO_4 is added & heated.	No dense white gas	Absence of Cl^-
4) A little of the salt soln is heated with conc. H_2SO_4 and paper ball is added.	No reddish brown fumes.	Absence of NO_3^-

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
5)	To the test soln add $BaCl_2$ soln.	white precipitate which is insoluble in dil. HCl.	Presence of sulphate (SO_4^{2-})

$$M_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2HCl$$

Confirmatory Tests

Sulphate

1)	To the test soln add lead acetate.	white precipitate	Presence of SO_4^{2-}

$$M_2SO_4 + Pb(CH_3COO)_2 \longrightarrow PbSO_4 + 2CH_3COOM$$

2)	To the above precipitate add ammonium acetate soln and heat it	white precipitate dissolves	Presence of sulphate confirmed.

$$PbSO_4 + 2CH_3COONH_4 \longrightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$$

Analysis of cation Intergroup separation

1)	To the test soln add	white precipitate	Absence of 0 group

2) To the above precipitate add ammonium acetate soln and heat it	white precipitate dissolves	Presence of sulphate confirmed.

$$PbSO_4 + 2CH_3COONH_4 \longrightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$$

Analysis of cation Intergroup seperation

1) To the test soln add dil Na_2CO_3	white precipitate	Absence of 0 group (NH_4^+)
2) To the test soln add dil. HCl	No white precipitate.	Absence of I group (Pb^{2+}).
3) To the test soln add dil. HCl, and pass H_2S gas.	No Black precipitate.	Absence of II group (Cu^{2+}).
4) To the test soln add dil HCl NH_4Cl & NH_4OH.	white gelatinous precipitate.	Presence of III group (Zn^{2+}, Mn^{2+}).
5) To the test soln add dil HCl NH_4Cl, NH_4OH Pass H_2S.	No white precipitate	Absence of IV group (Zn^{2+}, Mn^{2+})
6) To the test soln add dil. HCl, NH_4Cl, NH_4OH & $(NH_4)_2CO_3$	No white precipitate	Absence of V group (Ca^{2+}, Ba^{2+}, Sr^{2+})
7) To the test soln add dil HCl, NH_4Cl, NH_4OH & Na_2HCO_4.	No white precipitate	Absence of VI group (Mg^{2+}).

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Confirmatory test for Aluminium ion (Al^{3+})		
1)	Sodium hydroxide soln. is added in drops to excess to the test soln.	A white precipitate is formed which dissolves in excess of NaOH.	Presence of Al^{3+}
	$Al^{3+} + 3NaOH \longrightarrow Al(OH)_3 + 3Na^+$ $Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$		
2)	Ash test. The test soln is mixed with a few drops of conc. HNO_3 and a little cobalt nitrate soln. A rolled filter paper is dipped in ~~to~~ the soln and burnt to ash.	Blue tinted ash	Presence of Al^{3+} confirmed

$$2Co(NO_3)_2 \xrightarrow{\Delta} 2CoO + 4NO_2 + O_2$$
$$Al_2O_3 + CaO \longrightarrow Al_2O_3 \cdot CaO \text{ (Blue)}$$

Result

	of NaOH.	

$$Al^{3+} + 3NaOH \longrightarrow Al(OH)_3 + 3Na^+$$
$$Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$$

2) Ash test. The test soln is mixed with a few drops of conc. HNO_3 and a little cobalt nitrate soln. A rolled filter paper is dipped in the soln and burnt to ash.	Blue tinted ash	Presence of Al^{3+} confirmed

$$2Co(NO_3)_2 \xrightarrow{\Delta} 2CoO + 4NO_2 + O_2$$
$$Al_2O_3 + CaO \longrightarrow Al_2O_3 \cdot CaO \text{ (Blue)}$$

Result

The given salt contain aluminium as cation & sulphate as anion. Therefore the given salt is $Al_2(SO_4)_3$.

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	Salt 7 Preliminary Test		
1)	Note the colour of the given salt	Colourless	Absence of Cu^{2+}, Ni^{2+}, Mn^{2+}, Fe^{2+}.
2)	Note the smell of the given salt.	No characteristic smell	Absence of NH_4^+, CH_3COO^-.
3)	Solubility in dil. HCl.	Soluble	Absence of Pb^{2+}
4)	Flame test A pinch of salt is made conc. HCl and the paste is introduced into the flame by using a glass rod.	Pale green flame	Presence of Ba^{2+}.
5)	Ash test	No characteristic	Absence of

3) Solubility in dil. HCl	Soluble	Absence of Pb^{2+}
4) Flame test A pinch of salt is made conc. HCl and the paste is introduced into the flame by using a glass rod.	Pale green flame	Presence of Ba^{2+}

5) Ash test The test solution is mixed with few drops of conc HNO_3 & a little cobalt nitrate soln. A rolled filter paper dipped into it is burnt to ash.	No characteristic colour noted in the ash.	Absence of Al^{3+}, Zn^{2+}, Mg^{2+}.

Analysis of Anions

1) To the test soln add dil. HCl.	No Brisk effervescence	Absence of carbonate (CO_3^{-2}).
2) The test soln is rubbed with dil. H_2SO_4.	No smell of vinegar	Absence of nitrate (CH_3COO^-).
3) To a little of the salt soln ~~about~~ 1ml of conc H_2SO_4 is added & heated.	Dense white gas which fumes with NH_4OH soln taken on a glass rod.	Presence of chloride (Cl^-)

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
	$MCl + H_2SO_4 \longrightarrow MHSO_4 + HCl$		
	$HCl + NH_4OH \longrightarrow NH_4Cl$		
4)	A little of the salt soln is heated with conc H_2SO_4 and paper ball is added	No reddish brown fumes.	Absence of nitrate (NO_3^-)
5)	To the test soln add $BaCl_2$ soln	No white precipitate	Absence of sulphate (SO_4^{2-})
	Confirmatory Test. Chloride		
i)	The test soln is heated with conc H_2SO_4 and a pinch of a manganese dioxide.	Greenish yellow gas with irritating smell	Presence of chloride
	$2MCl + MnO_2 + 3H_2SO_4 \longrightarrow 2MHSO_4 + MnSO_4 + Cl_2 + 2H_2O$		
2)	The test soln is mixed	white curdy	Presence of chloride confirmed

1) The test soln is heated with conc H_2SO_4 and a pinch of a manganese dioxide.	Greenish yellow gas with irritating smell	Presence of chloride

$2HCl + MnO_2 + 3H_2SO_4 \longrightarrow 2KHSO_4 + MnSO_4 + Cl_2 + 2H_2O$

2) The test soln is mixed with dil HNO_3 and add $AgNO_3$ soln	white curdy precipitate soluble in excess of NH_4OH.	Presence of chloride confirmed.

$HCl + AgNO_3 \longrightarrow AgCl + HNO_3$

$AgCl + 2NH_4OH \longrightarrow [Ag(NH_3)_2]Cl + 2H_2O$

Analysis of cations Intergroup Separation

1) To the test soln add dil NaOH	white precipitate.	Absence of 0 group (NH_4^+)
2) To the test soln add dil-HCl	No white precipitate	Absence of I group (Pb^{2+})
3) To the test soln add dil. HCl, pass H_2S gas.	No Black precipitate.	Absence of II group (Cu^{2+})
4) To the test soln add dil. HCl, NH_4Cl and NH_4OH	No white gelatinous precipitate	Absence of III group (Al^{3+}, Fe^{2+})

SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
5)	To the test soln add dil HCl, NH_4Cl, NH_4OH & pass H_2S	No white or flesh colour precipitate	Absence of IV group (Zn^{2+}, Mn^{2+})
6)	To the test soln add dil HCl, NH_4Cl, NH_4OH & $(NH_4)_2CO_3$	white precipitate.	Absence of V group (Ca^{2+}, Ba^{2+}, Sr^{2+})
7)	To the test soln add dil HCl, NH_4Cl, NH_4OH & Na_2HPO_4	No white precipitate.	Absence of VI group (Mg^{2+}).

Confirmatory test for fifth group

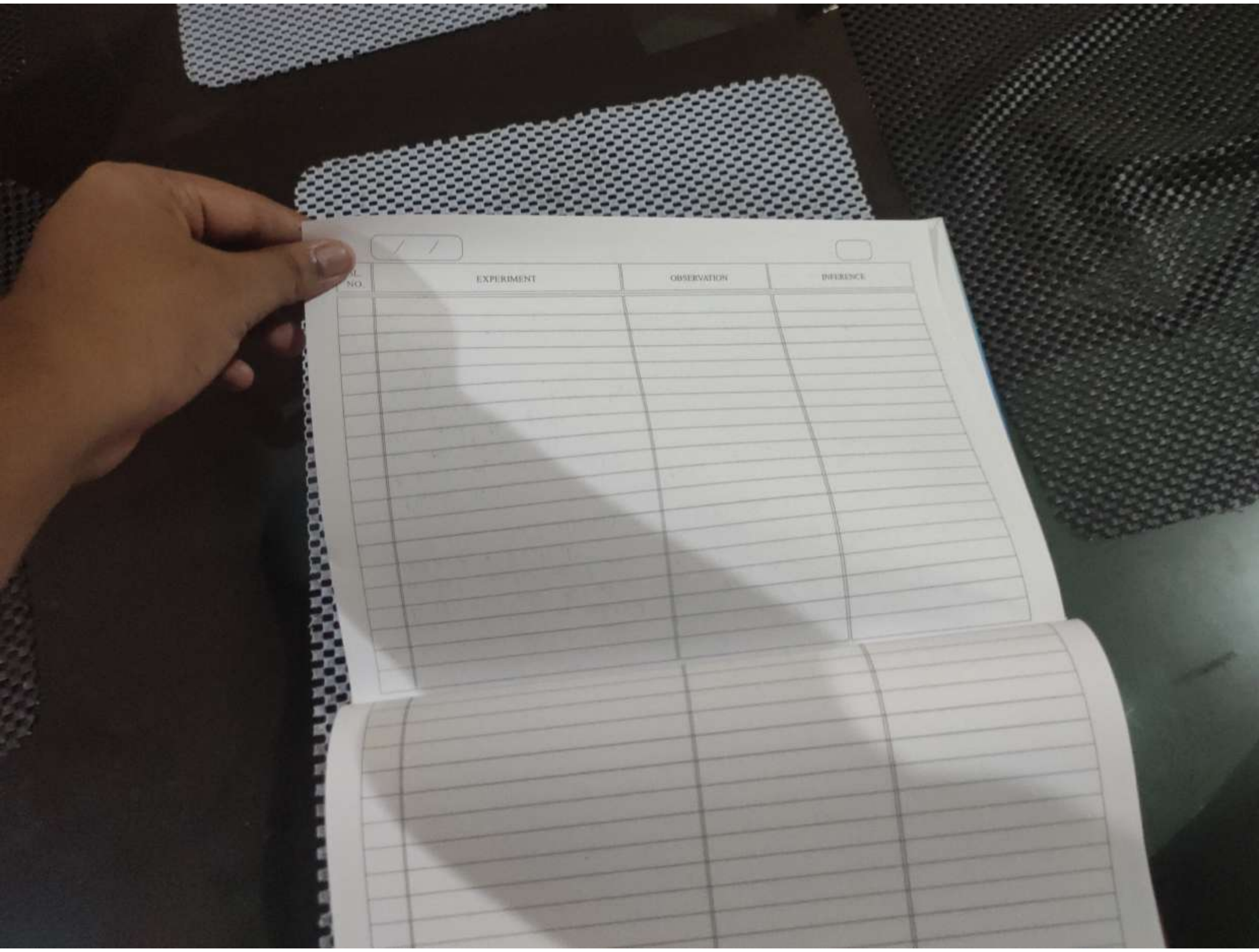
SL. NO.	EXPERIMENT	OBSERVATION	INFERENCE
1)	To the test solution acetic acid and pottasium chromate soln is added.	Yellow precipitate.	Presence of Ba^{2+}.
	$Ba^{2+} + K_2CrO_4 \longrightarrow$	$BaCrO_4 + 2K^+$	
2)	Ammonium oxalate soln & NH_4OH soln is added to the test soln The above	white precipitate readily dissolved in acetic acid.	Presence of Ba^{2+} confirmed.

1) To the test solution acetic acid and potassium chromate soln is added.	Yellow precipitate.	Presence of Ba^{2+}.
$Ba^{2+} + K_2CrO_4 \longrightarrow BaCrO_4 + 2K^+$		
2) Ammonium oxalate soln & NH_4OH soln is added to the test soln. The above precipitate acetic acid is added & heated.	White precipitate readily dissolved in acetic acid. Precipitate readily dissolved	Presence of Ba^{2+} confirmed. Presence of Ba^{2+}.
$Ba^{2+} + (NH_4)_2C_2O_4 \longrightarrow BaC_2O_4 + 2NH_4^+$		
3) Flame test A pinch of salt is made in the paste with conc. HCl and the paste is introduced in the flame by using a glass rod.	Pale green flame.	Presence of Ba^{2+}.

Result

The given contains Barium as cation and chloride as anion. Therefore the given salt is $BaCl_2$.

25/05/26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